Recitation 10 (Section 010)

Pranay Gundam

November 11, 2026

University of Minnesota

Please scan the QR code and fill out the attendance form!



Today's Agenda

- Announcements
- Midterm Review
  - Externalities
  - Trade between individuals
  - International Trade
  - Public/Private/Common Goods
  - Consumer Theory
  - Production Theory
- ICA

Announcements: Midterm 2

- On **Monday, November 16th**.
- My exam review session is this **Friday (11/13) at 2:30pm**.
- REMEMBER TO SHOW YOUR SCRATCH PIECE OF PAPER AND ID TO THE CAMERA.
- If you find any typos or errors in the exam, please continue to work and let us know after the exam. We'll check on this and follow up with you.

Externalities

Midterm 2 Review: Externalities

Externality: an uncompensated effect of someone's action on a bystander.

- Supply = **private** marginal cost (PMC). Demand = **private** marginal benefit (PMB).
- Social marginal cost: SMC = PMC + external marginal cost
- Social marginal benefit: SMB = PMB + external marginal benefit

	Market equilibrium	**Socially efficient**
Where	Supply = Demand	SMC = SMB
Negative ext.	Too much produced	Fix: Pigouvian **tax**
Positive ext.	Too little produced	Fix: Pigouvian **subsidy**

Recipe: (1) Find where the *private* curves cross = market. (2) Shift the curve that has the externality, find where the *social* curves cross = efficient. (3) Tax/subsidy = size of the external cost/benefit per unit.

Definitions check

[Vote] Negative or positive? On production or consumption?

1. A factory dumps waste into a river used by a fishery.
2. Your neighbor gets a flu shot.
3. Someone plays loud music on the bus.
4. A beekeeper's bees pollinate a nearby orchard.

Definitions check

[Vote] Negative or positive? On production or consumption?

1. A factory dumps waste into a river used by a fishery.
2. Your neighbor gets a flu shot.
3. Someone plays loud music on the bus.
4. A beekeeper's bees pollinate a nearby orchard.

Answer: 1. negative, production 2. positive, consumption 3. negative, consumption 4. positive, production.

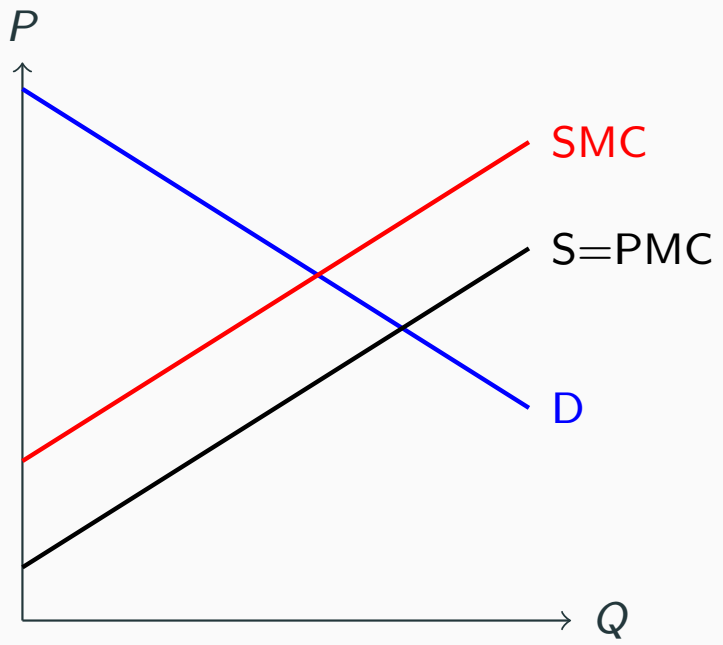
Which curve moves? Production externalities shift **cost** (SMC $\neq$ supply); consumption externalities shift **benefit** (SMB $\neq$ demand).

Problem: a negative externality

Demand: $P = 20 - Q$. Supply (PMC): $P = 2 + Q$.

Each unit causes \$4 of pollution damage.

1. Write the SMC curve.
2. Market quantity and price?
3. Socially efficient quantity?
4. Optimal Pigouvian tax?
5. Deadweight loss of the free market?



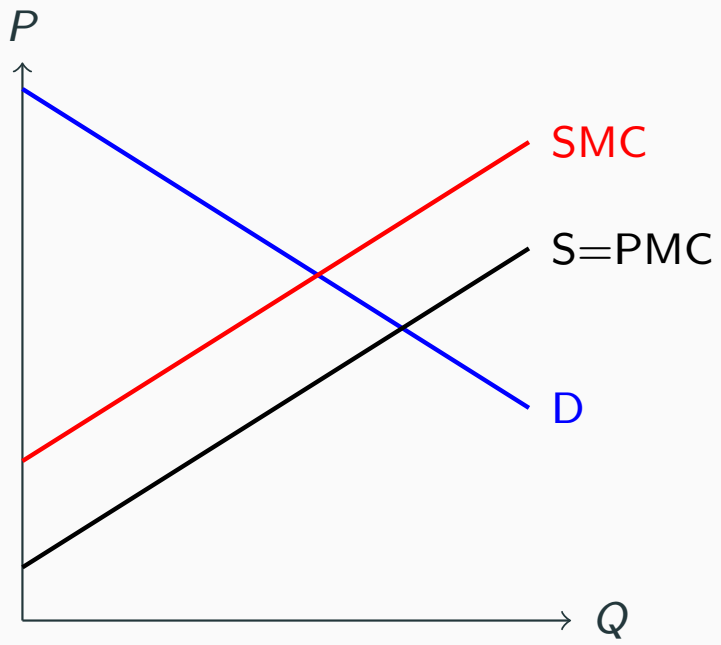
Problem: a negative externality

Demand: $P = 20 - Q$. Supply (PMC): $P = 2 + Q$.

Each unit causes \$4 of pollution damage.

1. Write the SMC curve.
2. Market quantity and price?
3. Socially efficient quantity?
4. Optimal Pigouvian tax?
5. Deadweight loss of the free market?

Answer: SMC: $P = 6 + Q$.

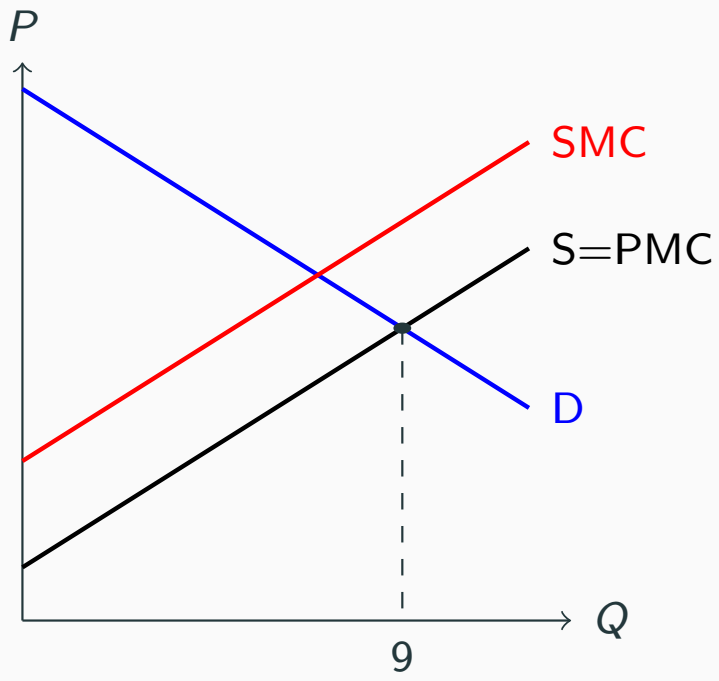


Problem: a negative externality

Demand: $P = 20 - Q$. Supply (PMC): $P = 2 + Q$.

Each unit causes \$4 of pollution damage.

1. Write the SMC curve.
2. Market quantity and price?
3. Socially efficient quantity?
4. Optimal Pigouvian tax?
5. Deadweight loss of the free market?



Answer: SMC: $P = 6 + Q$.

Answer:

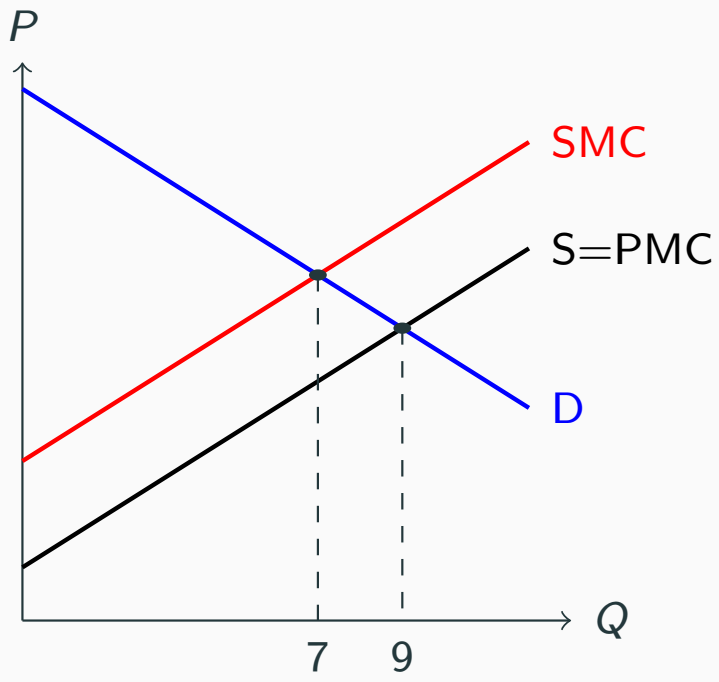
$20 - Q = 2 + Q \Rightarrow Q = 9,\ P = 11.$

Problem: a negative externality

Demand: $P = 20 - Q$. Supply (PMC): $P = 2 + Q$.

Each unit causes \$4 of pollution damage.

1. Write the SMC curve.
2. Market quantity and price?
3. Socially efficient quantity?
4. Optimal Pigouvian tax?
5. Deadweight loss of the free market?



Answer: SMC: $P = 6 + Q$.

Answer:

$20 - Q = 2 + Q \Rightarrow Q = 9,\ P = 11.$

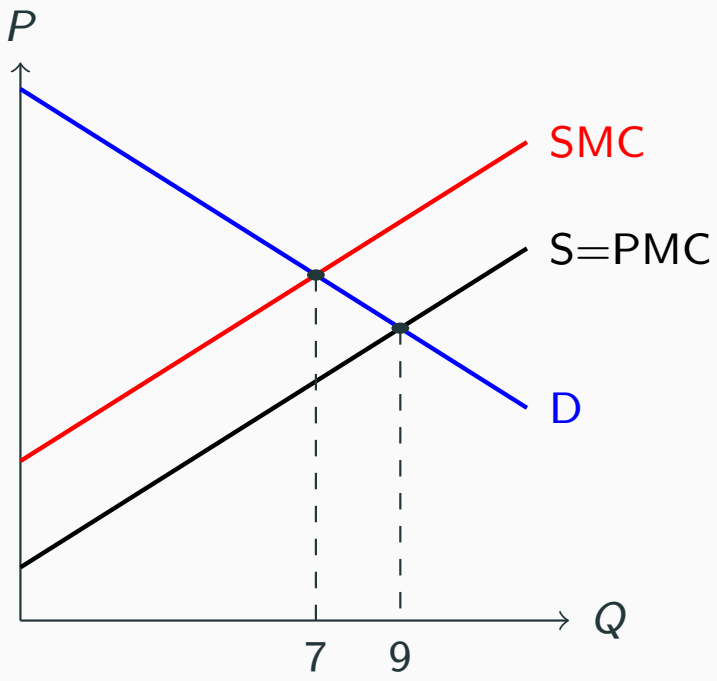
Answer: $20 - Q = 6 + Q \Rightarrow Q = 7.$

Problem: a negative externality

Demand: $P = 20 - Q$. Supply (PMC): $P = 2 + Q$.

Each unit causes \$4 of pollution damage.

1. Write the SMC curve.
2. Market quantity and price?
3. Socially efficient quantity?
4. Optimal Pigouvian tax?
5. Deadweight loss of the free market?



Answer: SMC: $P = 6 + Q$.

Answer:

$20 - Q = 2 + Q \Rightarrow Q = 9,\ P = 11.$

Answer: $20 - Q = 6 + Q \Rightarrow Q = 7.$

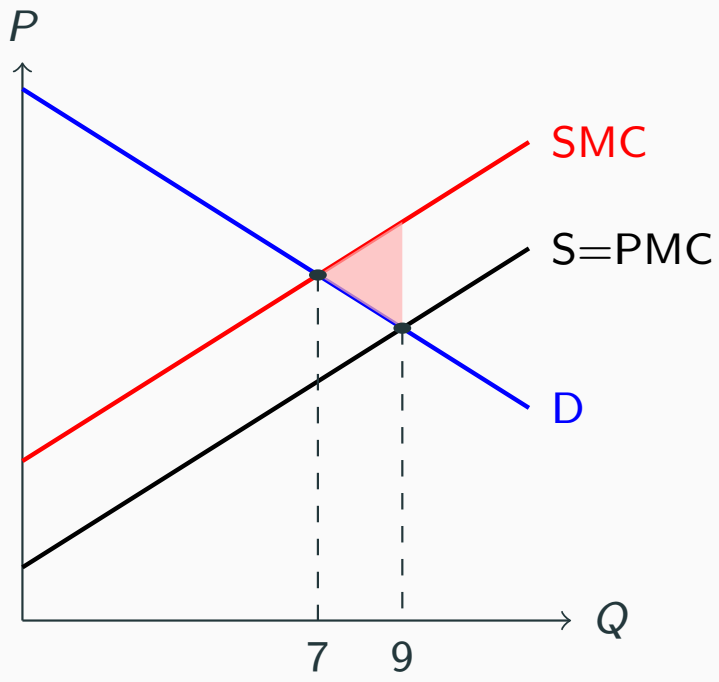
Answer: Tax = \$4 per unit.

Problem: a negative externality

Demand: $P = 20 - Q$. Supply (PMC): $P = 2 + Q$.

Each unit causes \$4 of pollution damage.

1. Write the SMC curve.
2. Market quantity and price?
3. Socially efficient quantity?
4. Optimal Pigouvian tax?
5. Deadweight loss of the free market?



Answer: SMC: $P = 6 + Q$.

Answer:

$20 - Q = 2 + Q \Rightarrow Q = 9,\ P = 11.$

Answer: $20 - Q = 6 + Q \Rightarrow Q = 7.$

Answer: Tax = \$4 per unit.

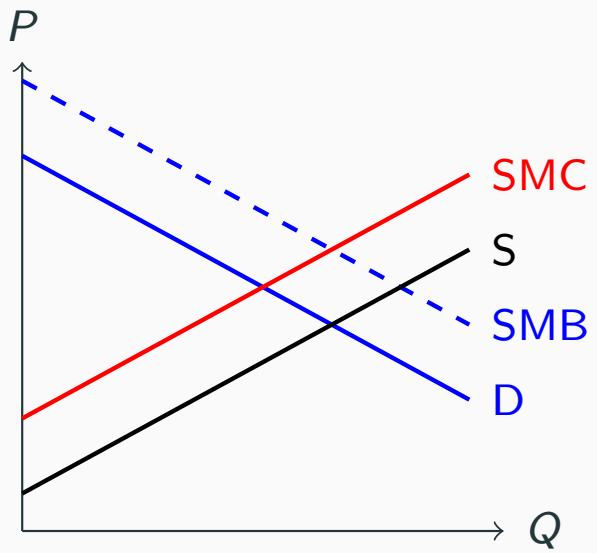
Answer: Red triangle:

$\frac{1}{2} \times (9 - 7) \times (15 - 11) = 4.$

Exam twist (2015): two externalities at once

Demand: $P = 20 - Q$. Supply: $P = 2 + Q$.
\$4 damage per unit produced: SMC $= 6 + Q$.
\$4 benefit to others per unit consumed: SMB $= 24 - Q$.

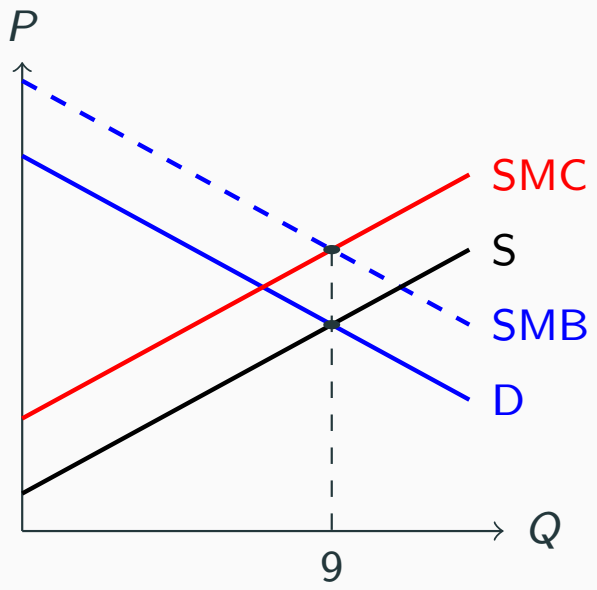
1. Market quantity? Efficient quantity?
2. Technology removes the pollution; the positive externality stays. New efficient quantity? Which policy?



Exam twist (2015): two externalities at once

Demand: $P = 20 - Q$. Supply: $P = 2 + Q$.
\$4 damage per unit produced: SMC $= 6 + Q$.
\$4 benefit to others per unit consumed: SMB $= 24 - Q$.

1. Market quantity? Efficient quantity?
2. Technology removes the pollution; the positive externality stays. New efficient quantity? Which policy?

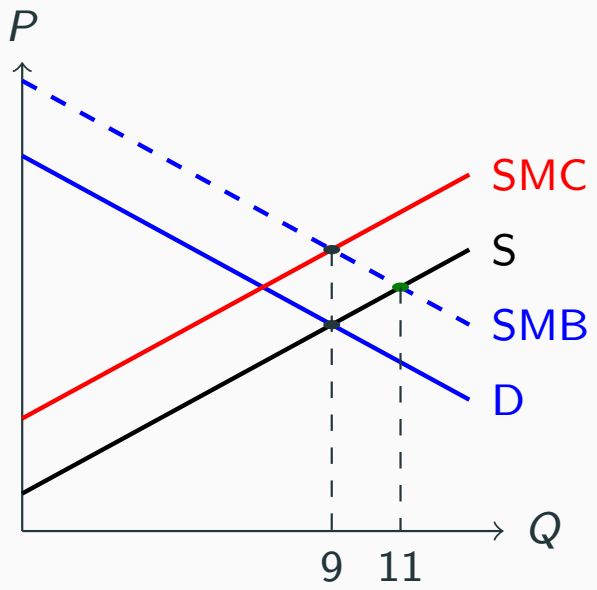


Answer: 1. Market: $20 - Q = 2 + Q \Rightarrow Q = 9$. Efficient: $24 - Q = 6 + Q \Rightarrow Q = 9$. The externalities cancel, so the market is already efficient (no tax, no subsidy).

Exam twist (2015): two externalities at once

Demand: $P = 20 - Q$. Supply: $P = 2 + Q$.
\$4 damage per unit produced: SMC $= 6 + Q$.
\$4 benefit to others per unit consumed: SMB $= 24 - Q$.

1. Market quantity? Efficient quantity?
2. Technology removes the pollution; the positive externality stays. New efficient quantity? Which policy?



Answer: 1. Market: $20 - Q = 2 + Q \Rightarrow Q = 9$. Efficient: $24 - Q = 6 + Q \Rightarrow Q = 9$. The externalities cancel, so the market is already efficient (no tax, no subsidy).
Answer: 2. Now SMC = S, so $24 - Q = 2 + Q \Rightarrow Q = 11$. Policy: a \$4 per-unit **subsidy** only.

Trade between individuals

Midterm 2 Review: Individual Trade

- **Absolute advantage:** can produce *more* of a good with the same inputs (more per hour).
- **Comparative advantage:** can produce a good at a *lower opportunity cost*.
- Opportunity cost of one good is the **inverse** of the opportunity cost of the other: if 1 cake costs 3 cookies, 1 cookie costs $\frac{1}{3}$ cake.

Which one decides who trades with whom?

Comparative advantage. Each person specializes in the good with the lower opportunity cost. Absolute advantage does not matter for gains from trade.

A person can have the absolute advantage in *both* goods, but never the comparative advantage in both (unless opportunity costs are equal, then no gains from trade).

Problem: Maya and Leo

Both work 5 hours a day.

Per hour	Cookies	Cakes
Maya	12	4
Leo	6	3

1. Who has the absolute advantage in cookies? In cakes?
2. Opportunity cost of 1 cake for each person?
3. Who has the comparative advantage in each good? Who exports what?
4. **[Vote]** Would "1 cake for 4 cookies" work as a price both accept?

Problem: Maya and Leo

Both work 5 hours a day.

Per hour	Cookies	Cakes
Maya	12	4
Leo	6	3

1. Who has the absolute advantage in cookies? In cakes?
2. Opportunity cost of 1 cake for each person?
3. Who has the comparative advantage in each good? Who exports what?
4. **[Vote]** Would "1 cake for 4 cookies" work as a price both accept?

Answer: Maya in both ($12 > 6$, $4 > 3$).

Problem: Maya and Leo

Both work 5 hours a day.

Per hour	Cookies	Cakes
Maya	12	4
Leo	6	3

1. Who has the absolute advantage in cookies? In cakes?
2. Opportunity cost of 1 cake for each person?
3. Who has the comparative advantage in each good? Who exports what?
4. **[Vote]** Would "1 cake for 4 cookies" work as a price both accept?

Answer: Maya in both ($12 > 6$, $4 > 3$).

Answer: Maya: 1 cake $=$ 3 cookies. Leo: 1 cake $=$ 2 cookies.

Problem: Maya and Leo

Both work 5 hours a day.

Per hour	Cookies	Cakes
Maya	12	4
Leo	6	3

1. Who has the absolute advantage in cookies? In cakes?
2. Opportunity cost of 1 cake for each person?
3. Who has the comparative advantage in each good? Who exports what?
4. **[Vote]** Would "1 cake for 4 cookies" work as a price both accept?

Answer: Maya in both ($12 > 6$, $4 > 3$).

Answer: Maya: 1 cake $=$ 3 cookies. Leo: 1 cake $=$ 2 cookies.

Answer: Leo has the CA in cakes ($2 < 3$); Maya in cookies ($\frac{1}{3} < \frac{1}{2}$ cake). Maya exports cookies, Leo exports cakes.

Problem: Maya and Leo

Both work 5 hours a day.

Per hour	Cookies	Cakes
Maya	12	4
Leo	6	3

1. Who has the absolute advantage in cookies? In cakes?
2. Opportunity cost of 1 cake for each person?
3. Who has the comparative advantage in each good? Who exports what?
4. **[Vote]** Would "1 cake for 4 cookies" work as a price both accept?

Answer: Maya in both ($12 > 6$, $4 > 3$).

Answer: Maya: 1 cake $=$ 3 cookies. Leo: 1 cake $=$ 2 cookies.

Answer: Leo has the CA in cakes ($2 < 3$); Maya in cookies ($\frac{1}{3} < \frac{1}{2}$ cake). Maya exports cookies, Leo exports cakes.

Answer: No. Maya can make a cake herself for 3 cookies, so she won't pay 4. Any price **between 2 and 3 cookies per cake** works.

Trade between individuals: PPF and true/false

PPFs (5 hours): Maya: 60 cookies *or* 20 cakes. Leo: 30 cookies *or* 15 cakes. The slope of the PPF is the opportunity cost. A flatter/steeper PPF = a different comparative advantage.

[Vote] True or false?

1. If Maya and Leo trade at 2.5 cookies per cake, both gain.
2. If both had the same PPF, there could be no gains from trade based on comparative advantage.
3. Leo invents a machine and now makes 10 cakes per hour. Maya still has the comparative advantage in cookies.

Trade between individuals: PPF and true/false

PPFs (5 hours): Maya: 60 cookies *or* 20 cakes. Leo: 30 cookies *or* 15 cakes. The slope of the PPF is the opportunity cost. A flatter/steeper PPF = a different comparative advantage.

[Vote] True or false?

1. If Maya and Leo trade at 2.5 cookies per cake, both gain.
2. If both had the same PPF, there could be no gains from trade based on comparative advantage.
3. Leo invents a machine and now makes 10 cakes per hour. Maya still has the comparative advantage in cookies.

Answer: True: the price is between the two opportunity costs.

Trade between individuals: PPF and true/false

PPFs (5 hours): Maya: 60 cookies *or* 20 cakes. Leo: 30 cookies *or* 15 cakes. The slope of the PPF is the opportunity cost. A flatter/steeper PPF = a different comparative advantage.

[Vote] True or false?

1. If Maya and Leo trade at 2.5 cookies per cake, both gain.
2. If both had the same PPF, there could be no gains from trade based on comparative advantage.
3. Leo invents a machine and now makes 10 cakes per hour. Maya still has the comparative advantage in cookies.

Answer: True: the price is between the two opportunity costs.
Answer: True (2015 exam). Gains from comparative advantage need *different* PPFs. (Increasing returns can still create gains: 2017 exam.)

Trade between individuals: PPF and true/false

PPFs (5 hours): Maya: 60 cookies *or* 20 cakes. Leo: 30 cookies *or* 15 cakes. The slope of the PPF is the opportunity cost. A flatter/steeper PPF = a different comparative advantage.

[Vote] True or false?

1. If Maya and Leo trade at 2.5 cookies per cake, both gain.
2. If both had the same PPF, there could be no gains from trade based on comparative advantage.
3. Leo invents a machine and now makes 10 cakes per hour. Maya still has the comparative advantage in cookies.

Answer: True: the price is between the two opportunity costs.
Answer: True (2015 exam). Gains from comparative advantage need *different* PPFs. (Increasing returns can still create gains: 2017 exam.)
Answer: True: Leo's cake now costs only 0.6 cookies, so his CA in cakes is even stronger.

International trade

Midterm 2 Review: International Trade

- Compare the **domestic (autarky) price** with the **world price**.
  - World price $<$ domestic price $\Rightarrow$ **import**. Consumers gain, producers lose.
  - World price $>$ domestic price $\Rightarrow$ **export**. Producers gain, consumers lose.
- With trade, domestic price = world price. Imports = $Q_D - Q_S$; exports = $Q_S - Q_D$.
- Either way, **total surplus rises**: winners gain more than losers lose.
- **Tariff:** raises the domestic price by the tariff. Consumers lose, producers gain, government gets revenue, and there is deadweight loss (two triangles).
- **Quota:** same price effect as the tariff; the "revenue" goes to whoever holds the import licenses.

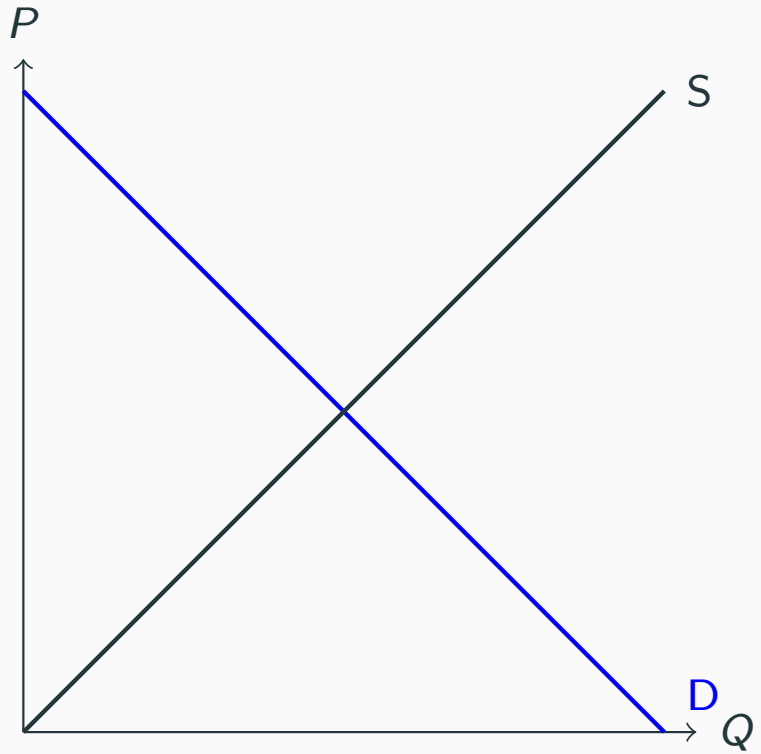
Worked example: setup (autarky)

Econland's market for widgets:

Demand: $P = 10 - Q$

Supply: $P = Q$

[Pair up] Find the autarky price and quantity, CS and PS.



Worked example: setup (autarky)

Econland's market for widgets:

Demand: $P = 10 - Q$

Supply: $P = Q$

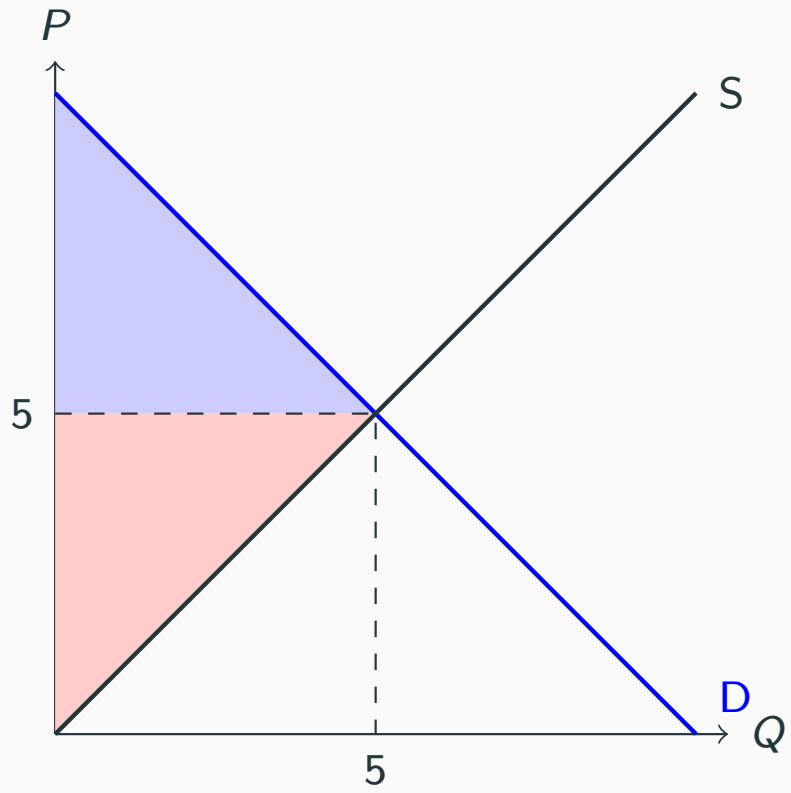
[Pair up] Find the autarky price and quantity, CS and PS.

Answer: $P = 5$, $Q = 5$.

$\text{CS} = \frac{1}{2} \times 5 \times 5 = 12.5$.

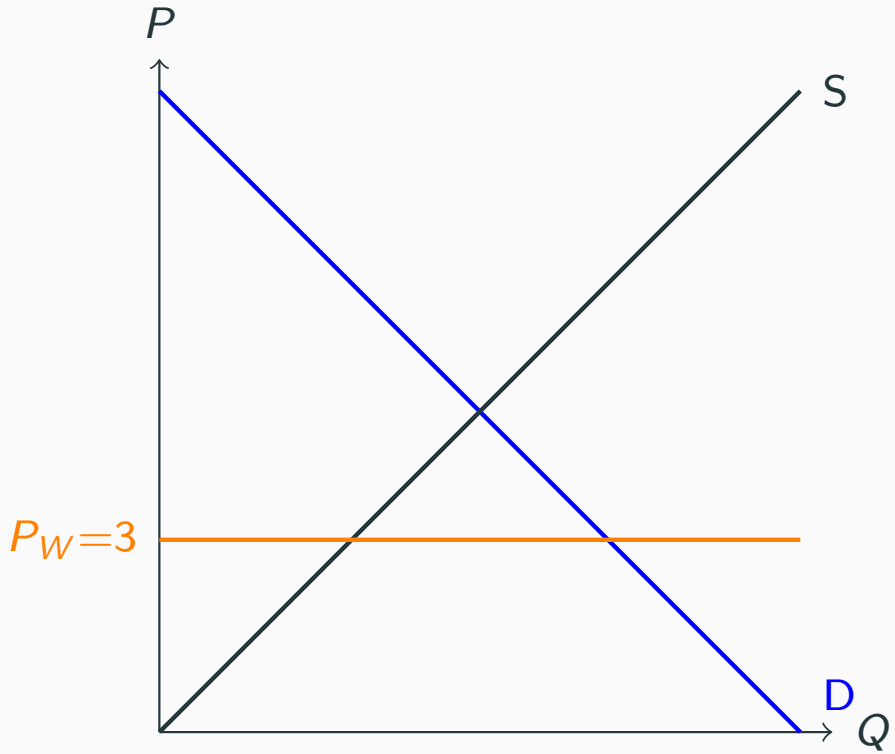
$\text{PS} = \frac{1}{2} \times 5 \times 5 = 12.5$.

Total surplus $= 25$.



Worked example 1: importing (world price = \$3)

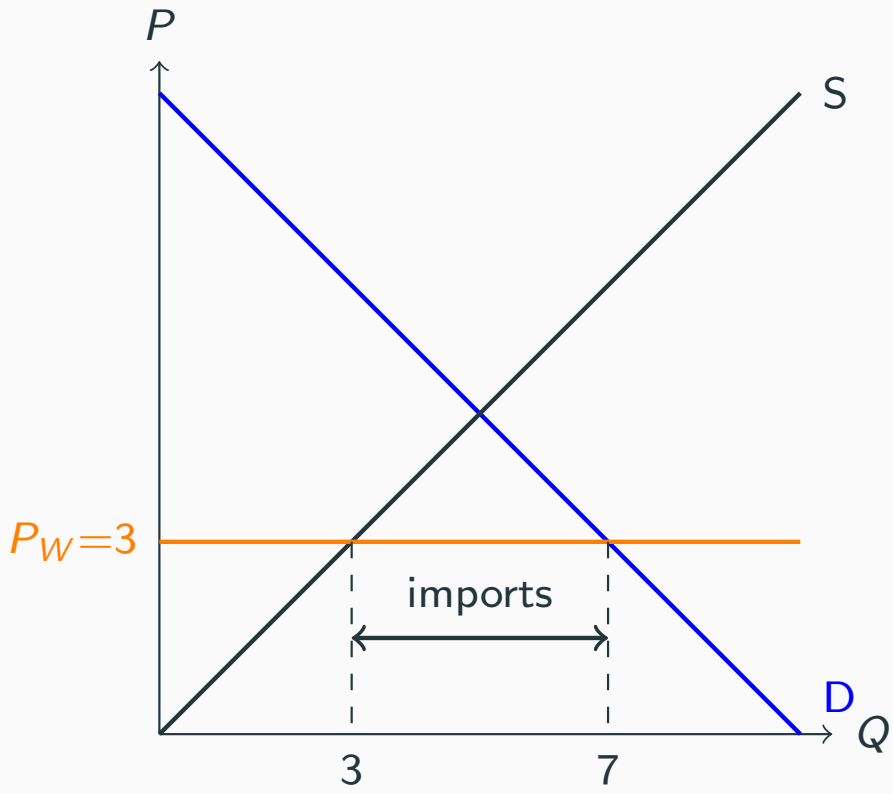
1. Import or export? How much?
2. New CS and PS? Who wins, who loses?
3. Change in total surplus?



Worked example 1: importing (world price = \$3)

1. Import or export? How much?
2. New CS and PS? Who wins, who loses?
3. Change in total surplus?

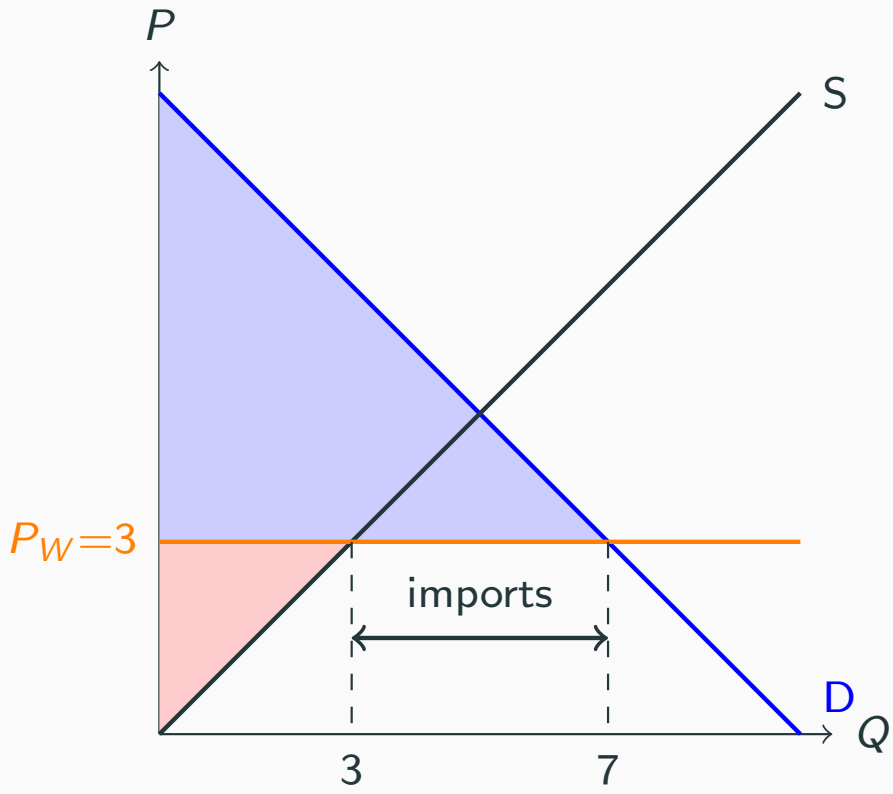
Answer: \$3 $<$ \$5, so import.
$Q_D = 7$, $Q_S = 3$, imports $= 4$.



Worked example 1: importing (world price = \$3)

1. Import or export? How much?
2. New CS and PS? Who wins, who loses?
3. Change in total surplus?

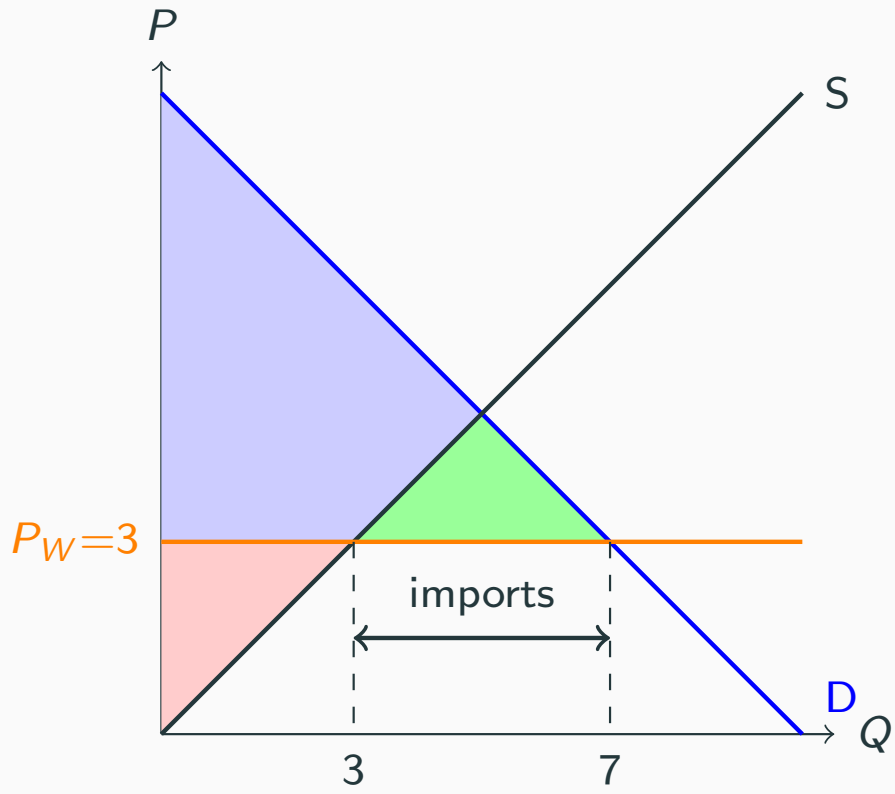
Answer: \$3 $<$ \$5, so import.
$Q_D = 7$, $Q_S = 3$, imports $= 4$.
Answer: CS $= \frac{1}{2} \cdot 7 \cdot 7 = 24.5$ (up 12). PS $= \frac{1}{2} \cdot 3 \cdot 3 = 4.5$ (down 8).
Consumers win, producers lose.



Worked example 1: importing (world price = \$3)

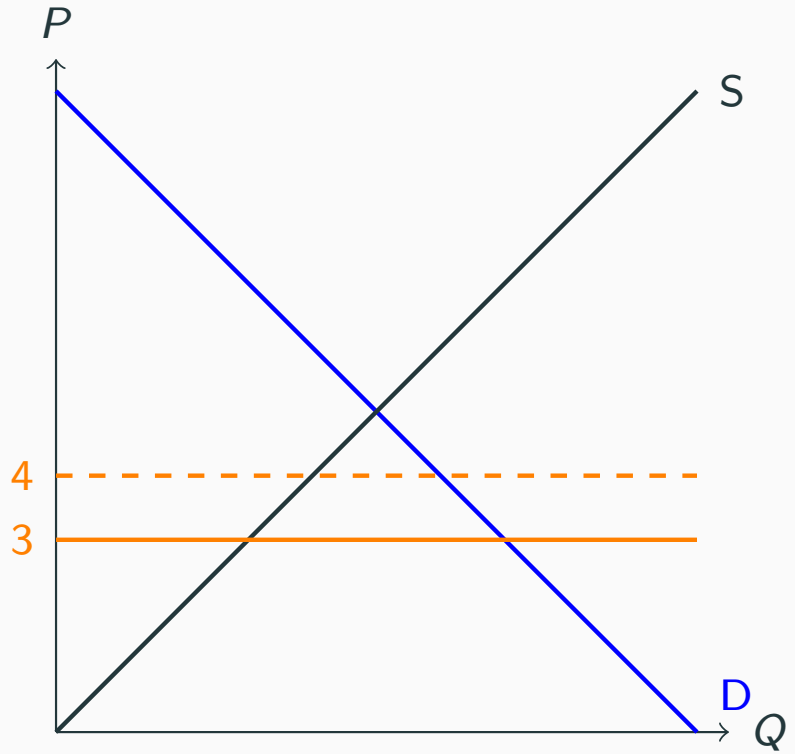
1. Import or export? How much?
2. New CS and PS? Who wins, who loses?
3. Change in total surplus?

Answer: \$3 < \$5, so import.
$Q_D = 7$, $Q_S = 3$, imports $= 4$.
Answer: CS $= \frac{1}{2} \cdot 7 \cdot 7 = 24.5$ (up 12). PS $= \frac{1}{2} \cdot 3 \cdot 3 = 4.5$ (down 8). Consumers win, producers lose.
Answer: $+4$: the green triangle, $\frac{1}{2} \times (7 - 3) \times (5 - 3)$.



Worked example 1 (cont.): a $1 tariff on imports

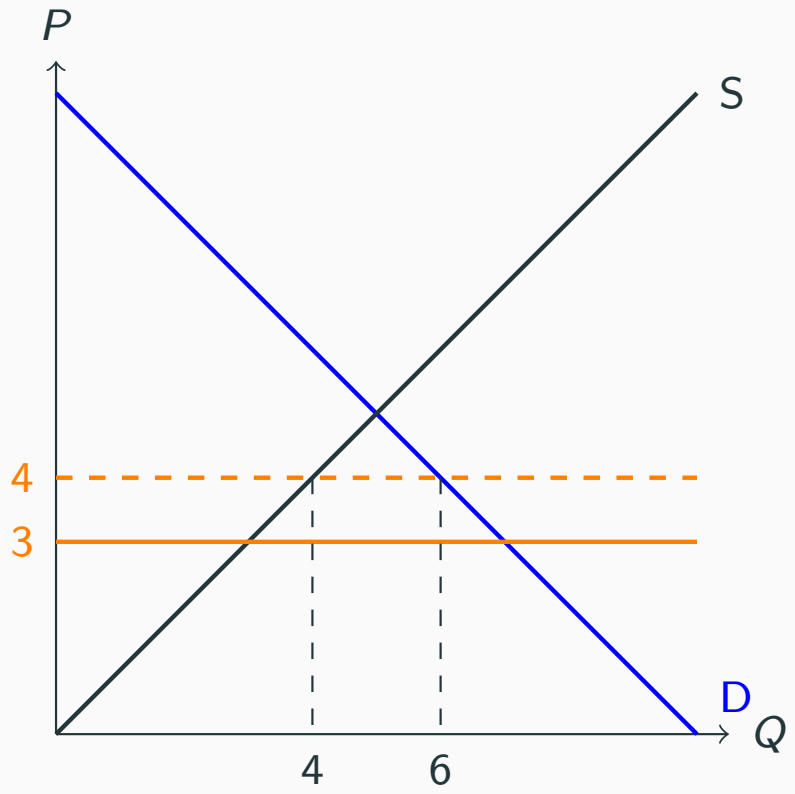
1. New domestic price? Imports?
2. Tariff revenue?
3. Deadweight loss compared with free trade?



Worked example 1 (cont.): a \$1 tariff on imports

1. New domestic price? Imports?
2. Tariff revenue?
3. Deadweight loss compared with free trade?

Answer: $P = 3 + 1 = 4$. $Q_D = 6$, $Q_S = 4$, imports $= 2$.

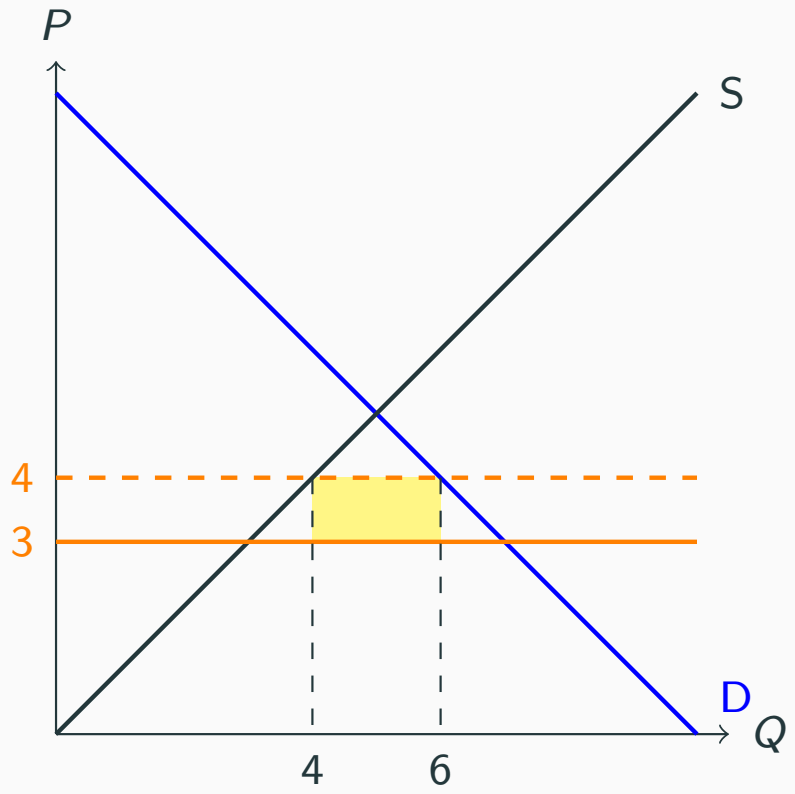


Worked example 1 (cont.): a \$1 tariff on imports

1. New domestic price? Imports?
2. Tariff revenue?
3. Deadweight loss compared with free trade?

Answer: $P = 3 + 1 = 4$. $Q_D = 6$, $Q_S = 4$, imports $= 2$.

Answer: \$1 $\times$ 2 = 2 (yellow box).



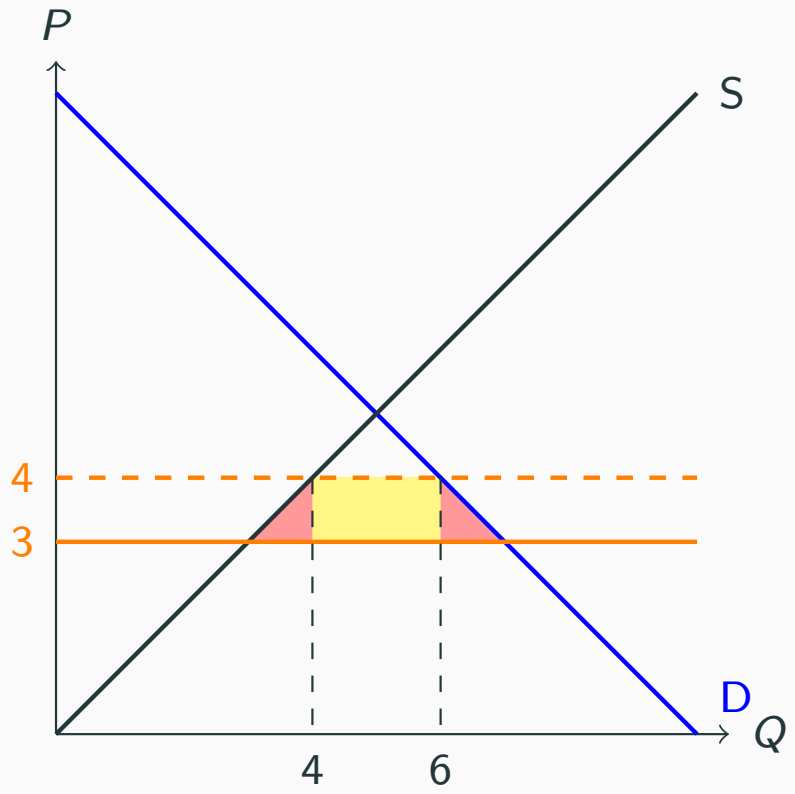
Worked example 1 (cont.): a \$1 tariff on imports

1. New domestic price? Imports?
2. Tariff revenue?
3. Deadweight loss compared with free trade?

Answer: $P = 3 + 1 = 4$. $Q_D = 6$, $Q_S = 4$, imports $= 2$.

Answer: \$1 $\times$ 2 $=$ 2 (yellow box).

Answer: Two triangles: $\frac{1}{2}(1)(1) + \frac{1}{2}(1)(1) = 1$. Check: free trade TS $= 29$; with tariff $18 + 8 + 2 = 28$.



Worked example 1 (cont.): a $1 tariff on imports

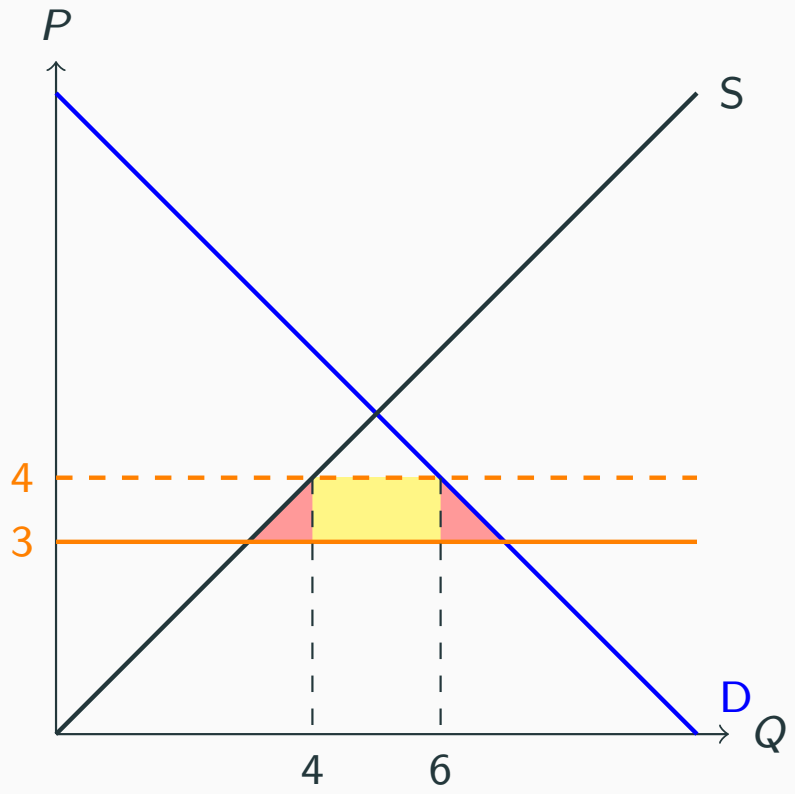
1. New domestic price? Imports?
2. Tariff revenue?
3. Deadweight loss compared with free trade?

Answer: $P = 3 + 1 = 4$. $Q_D = 6$, $Q_S = 4$, imports $= 2$.

Answer: $\$1 \times 2 = 2$ (yellow box).

Answer: Two triangles: $\frac{1}{2}(1)(1) + \frac{1}{2}(1)(1) = 1$. Check: free trade TS $= 29$; with tariff $18 + 8 + 2 = 28$.

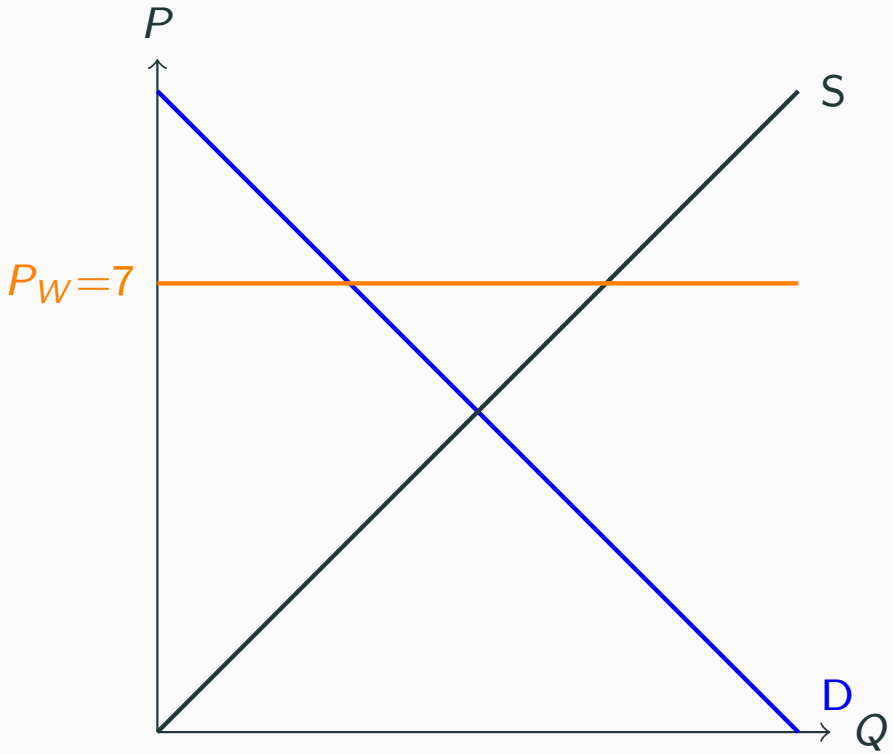
[Vote] A **quota of 2** imports instead: what is the domestic price? $4, the same as the tariff.



Worked example 2: exporting (world price = \$7)

Same market, but now the world price is \$7.

1. Import or export? How much?
2. New CS and PS? Who wins, who loses?
3. Change in total surplus?

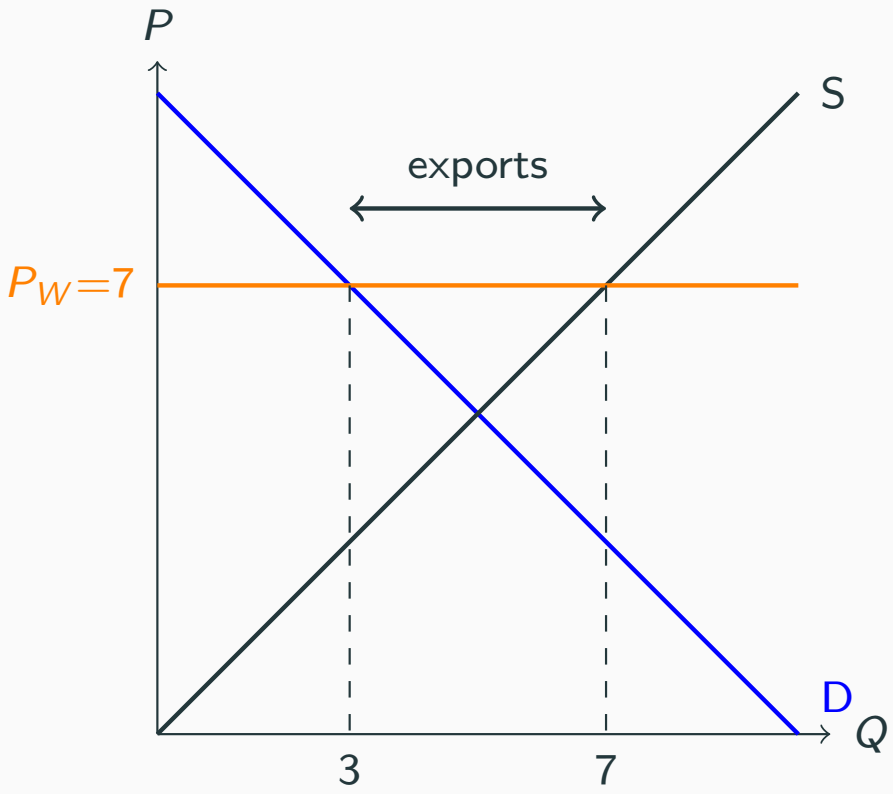


Worked example 2: exporting (world price = \$7)

Same market, but now the world price is \$7.

1. Import or export? How much?
2. New CS and PS? Who wins, who loses?
3. Change in total surplus?

Answer: \$7 $>$ \$5, so export. $Q_S = 7$, $Q_D = 3$, exports $= 4$.



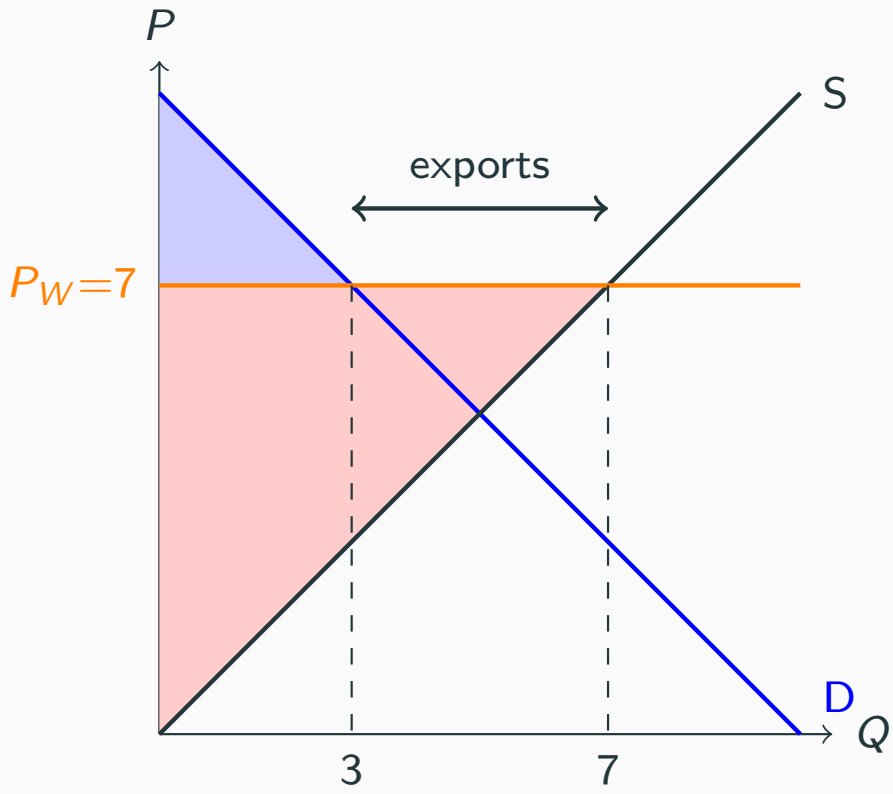
Worked example 2: exporting (world price = \$7)

Same market, but now the world price is \$7.

1. Import or export? How much?
2. New CS and PS? Who wins, who loses?
3. Change in total surplus?

Answer: \$7 > \$5, so export. $Q_S = 7$, $Q_D = 3$, exports $= 4$.

Answer: CS $= \frac{1}{2} \cdot 3 \cdot 3 = 4.5$ (down 8). PS $= \frac{1}{2} \cdot 7 \cdot 7 = 24.5$ (up 12). Producers win, consumers lose.



Worked example 2: exporting (world price = \$7)

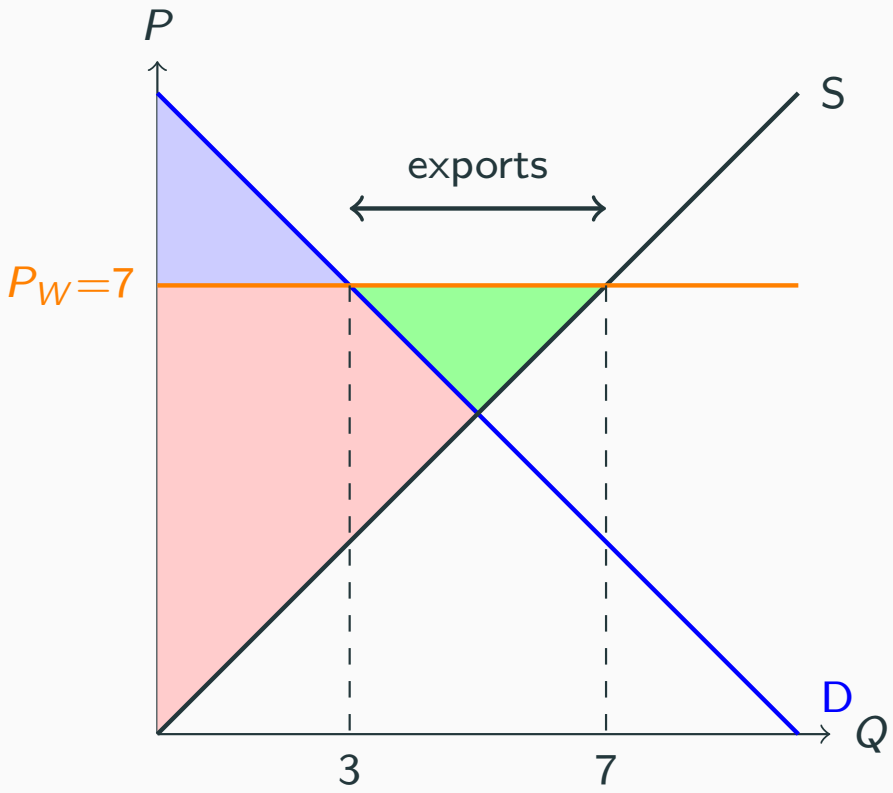
Same market, but now the world price is \$7.

1. Import or export? How much?
2. New CS and PS? Who wins, who loses?
3. Change in total surplus?

Answer: \$7 > \$5, so export. $Q_S = 7$, $Q_D = 3$, exports $= 4$.

Answer: CS $= \frac{1}{2} \cdot 3 \cdot 3 = 4.5$ (down 8). PS $= \frac{1}{2} \cdot 7 \cdot 7 = 24.5$ (up 12). Producers win, consumers lose.

Answer: $+4$ again: $\frac{1}{2} \times (7 - 3) \times (7 - 5)$.



Public, private and common goods

Midterm 2 Review: Public/Private/Common Goods

- **Excludable:** people who don't pay can be kept from using it.
- **Rival:** one person's use leaves less for others.

	Rival	**Non-rival**
Excludable	Private good (pizza)	Club good (Netflix)
Non-excl.	Common resource (ocean fish)	Public good (defense)

- **Public goods:** free-rider problem, so markets under-provide. Efficient to provide if **sum of everyone's willingness to pay** $\geq$ **cost**.
- **Common resources:** tragedy of the commons, so they are overused (a negative externality). Fixes: tax, regulate, or make it private. Asking people to cut back voluntarily does *not* work.

Definitions check: classify it

[Vote] Private, club, common resource, or public good?

1. A banana
2. FM radio
3. Congested non-toll road
4. Elephants roaming freely in Kenya
5. HBO / premium TV channel
6. A fireworks show in a public park
7. Timber from an unregulated forest
8. An uncongested toll road

Definitions check: classify it

[Vote] Private, club, common resource, or public good?

1. A banana
2. FM radio
3. Congested non-toll road
4. Elephants roaming freely in Kenya
5. HBO / premium TV channel
6. A fireworks show in a public park
7. Timber from an unregulated forest
8. An uncongested toll road

Answer: 1. private 2. public 3. common 4. common 5. club 6. public 7. common 8. club.

Exam pattern: "Which are *rival*?" $\Rightarrow$ private goods and common resources (1, 3, 4, 7).

Problem: the town fireworks show

Resident	WTP ($)
Ana	9
Ben	7
Cam	5
Dee	3
Eli	1

The show is non-rival and non-excludable. It costs $20.

1. Is it socially efficient to hold the show?
2. Will a private firm hold it? Why or why not?
3. The show moves to a private island where tickets can be checked (excludable). Can the owner break even with one ticket price?

Problem: the town fireworks show

Resident	WTP ($)
Ana	9
Ben	7
Cam	5
Dee	3
Eli	1

The show is non-rival and non-excludable. It costs $20.

1. Is it socially efficient to hold the show?
2. Will a private firm hold it? Why or why not?
3. The show moves to a private island where tickets can be checked (excludable). Can the owner break even with one ticket price?

Answer: Yes: total WTP $= 9 + 7 + 5 + 3 + 1 = 25 \geq 20$.

Problem: the town fireworks show

Resident	WTP ($)
Ana	9
Ben	7
Cam	5
Dee	3
Eli	1

The show is non-rival and non-excludable. It costs $20.

1. Is it socially efficient to hold the show?
2. Will a private firm hold it? Why or why not?
3. The show moves to a private island where tickets can be checked (excludable). Can the owner break even with one ticket price?

Answer: Yes: total WTP $= 9 + 7 + 5 + 3 + 1 = 25 \geq 20$.

Answer: No: non-excludable, so people free-ride and nobody pays. The government could tax each person $4 and hold it.

Problem: the town fireworks show

Resident	WTP ($)
Ana	9
Ben	7
Cam	5
Dee	3
Eli	1

The show is non-rival and non-excludable. It costs $20.

1. Is it socially efficient to hold the show?
2. Will a private firm hold it? Why or why not?
3. The show moves to a private island where tickets can be checked (excludable). Can the owner break even with one ticket price?

Answer: Yes: total WTP $= 9 + 7 + 5 + 3 + 1 = 25 \geq 20$.

Answer: No: non-excludable, so people free-ride and nobody pays. The government could tax each person $4 and hold it.

Answer: No. Best revenue: $9 (1 buyer) $= 9$, $7 (2) $= 14$, $5 (3) $= 15$, $3 (4) $= 12$, $1 (5) $= 5$. Max is $15 $< 20, even though the show is efficient.

Consumer theory

Midterm 2 Review: Consumer Theory

- **Budget constraint:** $P_X X + P_Y Y = I$. Slope $= -P_X/P_Y$ (the relative price).
- **Indifference curve:** bundles the consumer likes equally. Slope = **MRS**.
- **Optimum:** tangency, MRS $= P_X/P_Y$.
- Special cases: **perfect substitutes** (straight ICs), **perfect complements** (L-shaped ICs).

Change	**Budget line**
Income rises	Shifts **out**, same slope. Normal goods: buy more. Inferior goods: buy less.
One price falls	**Rotates** out along that good's axis; slope changes.

Income vs. substitution effects

When the price of X falls:

- **Substitution effect:** X is relatively cheaper, so buy **more** X. Always this direction. Move *along* the original IC.
- **Income effect:** purchasing power rises. Buy more X if normal, less if inferior. Move to a *new* IC.

P_X falls	Substitution	Income	Total
Normal good	$+$	$+$	$+$
Inferior good	$+$	$-$ (smaller)	$+$
Giffen good	$+$	$-$ (**larger**)	$-$

[Vote] A good's price falls and people buy *less* of it. What kind of good? Which effect is bigger?

Income vs. substitution effects

When the price of X falls:

- **Substitution effect:** X is relatively cheaper, so buy **more** X. Always this direction. Move *along* the original IC.
- **Income effect:** purchasing power rises. Buy more X if normal, less if inferior. Move to a *new* IC.

P_X falls	Substitution	Income	Total
Normal good	$+$	$+$	$+$
Inferior good	$+$	$-$ (smaller)	$+$
Giffen good	$+$	$-$ (**larger**)	$-$

[Vote] A good's price falls and people buy *less* of it. What kind of good? Which effect is bigger?

Answer: A Giffen good: the income effect goes the opposite way and is larger than the substitution effect.

Problem: budget constraints

Jamie has \$40 and buys apples (A, horizontal axis) at \$4 and bananas (B, vertical axis) at \$2.

1. Intercepts of the budget line? Opportunity cost of 1 apple?
2. Income doubles to \$80. What happens to the line?
3. Back to \$40, but apples fall to \$2. What happens to the line?
4. At the optimum, MRS (bananas per apple) equals what at the original prices?

Problem: budget constraints

Jamie has \$40 and buys apples (A, horizontal axis) at \$4 and bananas (B, vertical axis) at \$2.

1. Intercepts of the budget line? Opportunity cost of 1 apple?
2. Income doubles to \$80. What happens to the line?
3. Back to \$40, but apples fall to \$2. What happens to the line?
4. At the optimum, MRS (bananas per apple) equals what at the original prices?

Answer: 10 apples or 20 bananas. 1 apple costs 2 bananas (slope $-4/2 = -2$).

Problem: budget constraints

Jamie has \$40 and buys apples (A, horizontal axis) at \$4 and bananas (B, vertical axis) at \$2.

1. Intercepts of the budget line? Opportunity cost of 1 apple?
2. Income doubles to \$80. What happens to the line?
3. Back to \$40, but apples fall to \$2. What happens to the line?
4. At the optimum, MRS (bananas per apple) equals what at the original prices?

Answer: 10 apples or 20 bananas. 1 apple costs 2 bananas (slope $-4/2 = -2$).

Answer: Parallel shift out: 20 apples or 40 bananas. Slope still -2.

Problem: budget constraints

Jamie has \$40 and buys apples (A, horizontal axis) at \$4 and bananas (B, vertical axis) at \$2.

1. Intercepts of the budget line? Opportunity cost of 1 apple?
2. Income doubles to \$80. What happens to the line?
3. Back to \$40, but apples fall to \$2. What happens to the line?
4. At the optimum, MRS (bananas per apple) equals what at the original prices?

Answer: 10 apples or 20 bananas. 1 apple costs 2 bananas (slope $-4/2 = -2$).
Answer: Parallel shift out: 20 apples or 40 bananas. Slope still -2.
Answer: Rotates out on the apple axis: 20 apples or 20 bananas. Slope -1.

Problem: budget constraints

Jamie has \$40 and buys apples (A, horizontal axis) at \$4 and bananas (B, vertical axis) at \$2.

1. Intercepts of the budget line? Opportunity cost of 1 apple?
2. Income doubles to \$80. What happens to the line?
3. Back to \$40, but apples fall to \$2. What happens to the line?
4. At the optimum, MRS (bananas per apple) equals what at the original prices?

Answer: 10 apples or 20 bananas. 1 apple costs 2 bananas (slope $-4/2 = -2$).
Answer: Parallel shift out: 20 apples or 40 bananas. Slope still -2.
Answer: Rotates out on the apple axis: 20 apples or 20 bananas. Slope -1.
Answer: MRS $= P_A/P_B = 2$.

Problem: perfect complements (2017 exam style)

Sam always eats hot dogs and buns together, 1 for 1. Income \$24. Each costs \$2.

1. What is Sam's optimal bundle?
2. The price of hot dogs falls to \$1. New bundle? Total change in hot dogs?
3. Split the change into substitution and income effects.

[Pair up] Why? Explain to your neighbor why the substitution effect is zero here but would be large for perfect substitutes.

Problem: perfect complements (2017 exam style)

Sam always eats hot dogs and buns together, 1 for 1. Income \$24. Each costs \$2.

1. What is Sam's optimal bundle?
2. The price of hot dogs falls to \$1. New bundle? Total change in hot dogs?
3. Split the change into substitution and income effects.

Answer: Each pair costs \$4, so 6 pairs: (6 hot dogs, 6 buns).

[Pair up] Why? Explain to your neighbor why the substitution effect is zero here but would be large for perfect substitutes.

Problem: perfect complements (2017 exam style)

Sam always eats hot dogs and buns together, 1 for 1. Income \$24. Each costs \$2.

1. What is Sam's optimal bundle?
2. The price of hot dogs falls to \$1. New bundle? Total change in hot dogs?
3. Split the change into substitution and income effects.

Answer: Each pair costs \$4, so 6 pairs: (6 hot dogs, 6 buns).
Answer: Each pair now costs \$3: 8 pairs. Hot dogs rise by 2.

[Pair up] Why? Explain to your neighbor why the substitution effect is zero here but would be large for perfect substitutes.

Problem: perfect complements (2017 exam style)

Sam always eats hot dogs and buns together, 1 for 1. Income \$24. Each costs \$2.

1. What is Sam's optimal bundle?
2. The price of hot dogs falls to \$1. New bundle? Total change in hot dogs?
3. Split the change into substitution and income effects.

Answer: Each pair costs \$4, so 6 pairs: (6 hot dogs, 6 buns).
Answer: Each pair now costs \$3: 8 pairs. Hot dogs rise by 2.
Answer: Substitution effect $= 0$: with L-shaped ICs, Sam never trades buns for hot dogs. Income effect $= +2$.

[Pair up] Why? Explain to your neighbor why the substitution effect is zero here but would be large for perfect substitutes.

Production theory

Midterm 2 Review: Production Theory – cost definitions

- $TC = FC + VC$
- $FC =$ cost at $Q = 0$
- $ATC = TC/Q, \;\; AVC = VC/Q$
- $AFC = FC/Q,$
  $ATC = AFC + AVC$
- $FC = (ATC - AVC) \times Q$
- $MC = \Delta TC/\Delta Q$
- Profit
  $= TR - TC = (P - ATC) \times Q$

- MC crosses ATC and AVC at their **minimum** points.
- Competitive firm:
  $P = MR = AR$.
- Choose Q where $P = MC$.
- The gap between ATC and AVC shrinks as Q grows (it is AFC).

[Pair up] Graph exam trick: to get fixed cost from the graph, pick any Q and compute $(ATC - AVC) \times Q$.

Midterm 2 Review: Production Theory – the firm's decisions

	Short run (FC is sunk)	**Long run** (all costs avoidable)
Rule	Shut down if $P < AVC$	Exit if $P < ATC$ Enter if $P > ATC$
Supply curve	MC above min AVC	MC above min ATC
Loss if stop	FC	0

Long-run equilibrium: entry/exit until profit $= 0$, so $P =$ **min ATC**.
Number of firms $= Q_{\text{market}}/q_{\text{firm}}$.

Long-run supply is perfectly flat if: (1) free entry and exit, (2) same technology for all firms, (3) input prices don't change as the industry grows.
(This question was on the 2015, 2017 *and* 2019 exams.)

Problem: one firm's costs

Q	TC	MC	ATC	AVC
0	10	–	–	–
1	14	?	?	?
2	16	?	?	?
3	20	?	?	?
4	26	?	?	?
5	34	?	?	?
6	44	?	?	?

[Pair up] Fill in the table. Then:

1. Fixed cost?
2. Minimum ATC and AVC? At what Q?
3. Below what price does the firm **shut down**?
4. What will the long-run price be?

Problem: one firm's costs

Q	TC	MC	ATC	AVC
0	10	–	–	–
1	14	4	14	4
2	16	2	8	3
3	20	4	6.67	3.33
4	26	6	6.5	4
5	34	8	6.8	4.8
6	44	10	7.33	5.67

[Pair up] Fill in the table. Then:

1. Fixed cost?
2. Minimum ATC and AVC? At what Q?
3. Below what price does the firm **shut down**?
4. What will the long-run price be?

Problem: one firm's costs

Q	TC	MC	ATC	AVC
0	10	–	–	–
1	14	4	14	4
2	16	2	8	3
3	20	4	6.67	3.33
4	26	6	6.5	4
5	34	8	6.8	4.8
6	44	10	7.33	5.67

[Pair up] Fill in the table. Then:

1. Fixed cost?
2. Minimum ATC and AVC? At what Q?
3. Below what price does the firm **shut down**?
4. What will the long-run price be?

Answer: $FC = 10$.

Problem: one firm's costs

Q	TC	MC	ATC	AVC
0	10	–	–	–
1	14	4	14	4
2	16	2	8	3
3	20	4	6.67	3.33
4	26	6	6.5	4
5	34	8	6.8	4.8
6	44	10	7.33	5.67

[Pair up] Fill in the table. Then:

1. Fixed cost?
2. Minimum ATC and AVC? At what Q?
3. Below what price does the firm **shut down**?
4. What will the long-run price be?

Answer: $FC = 10$.

Answer: Min ATC $= 6.5$ at $Q = 4$; min AVC $= 3$ at $Q = 2$.

Problem: one firm's costs

Q	TC	MC	ATC	AVC
0	10	–	–	–
1	14	4	14	4
2	16	2	8	3
3	20	4	6.67	3.33
4	26	6	6.5	4
5	34	8	6.8	4.8
6	44	10	7.33	5.67

[Pair up] Fill in the table. Then:

1. Fixed cost?
2. Minimum ATC and AVC? At what Q?
3. Below what price does the firm **shut down**?
4. What will the long-run price be?

Answer: $FC = 10$.
Answer: Min ATC $= 6.5$ at $Q = 4$; min AVC $= 3$ at $Q = 2$.
Answer: Below \$3 (min AVC).

Problem: one firm's costs

Q	TC	MC	ATC	AVC
0	10	–	–	–
1	14	4	14	4
2	16	2	8	3
3	20	4	6.67	3.33
4	26	6	6.5	4
5	34	8	6.8	4.8
6	44	10	7.33	5.67

[Pair up] Fill in the table. Then:

1. Fixed cost?
2. Minimum ATC and AVC? At what Q?
3. Below what price does the firm **shut down**?
4. What will the long-run price be?

Answer: $FC = 10$.
Answer: Min ATC $= 6.5$ at $Q = 4$; min AVC $= 3$ at $Q = 2$.
Answer: Below \$3 (min AVC).
Answer: \$6.50 (min ATC), each firm makes 4 units.

Problem: same firm, different prices

Use the cost table from the last slide. Produce every unit whose $MC \leq P$.

1. $P = \$9$: how many units? Profit?
2. $P = \$5$: how many units? Profit? Produce or shut down in the short run? Long run?
3. $P = \$2.50$: what does the firm do?
4. In the long run $P = \$6.50$. Market demand is $Q_D = 1400 - 80P$. How many firms?

Problem: same firm, different prices

Use the cost table from the last slide. Produce every unit whose $MC \leq P$.

1. $P = \$9$: how many units? Profit?
2. $P = \$5$: how many units? Profit? Produce or shut down in the short run? Long run?
3. $P = \$2.50$: what does the firm do?
4. In the long run $P = \$6.50$. Market demand is $Q_D = 1400 - 80P$. How many firms?

Answer: $Q = 5$ (MC of 5th unit $= 8 \leq 9$; 6th $= 10 > 9$). Profit $= 45 - 34 = 11 = (9 - 6.8) \times 5$.

Problem: same firm, different prices

Use the cost table from the last slide. Produce every unit whose $MC \leq P$.

1. $P = \$9$: how many units? Profit?
2. $P = \$5$: how many units? Profit? Produce or shut down in the short run? Long run?
3. $P = \$2.50$: what does the firm do?
4. In the long run $P = \$6.50$. Market demand is $Q_D = 1400 - 80P$. How many firms?

Answer: $Q = 5$ (MC of 5th unit $= 8 \leq 9$; 6th $= 10 > 9$). Profit $= 45 - 34 = 11 = (9 - 6.8) \times 5$.

Answer: $Q = 3$. Profit $= 15 - 20 = -5$. Shutting down loses $FC = 10$, so produce ($P = 5 > AVC = 3.33$). Long run: exit ($P < ATC$).

Problem: same firm, different prices

Use the cost table from the last slide. Produce every unit whose $MC \leq P$.

1. $P = \$9$: how many units? Profit?
2. $P = \$5$: how many units? Profit? Produce or shut down in the short run? Long run?
3. $P = \$2.50$: what does the firm do?
4. In the long run $P = \$6.50$. Market demand is $Q_D = 1400 - 80P$. How many firms?

Answer: $Q = 5$ (MC of 5th unit $= 8 \leq 9$; 6th $= 10 > 9$). Profit $= 45 - 34 = 11 = (9 - 6.8) \times 5$.
Answer: $Q = 3$. Profit $= 15 - 20 = -5$. Shutting down loses $FC = 10$, so produce ($P = 5 > AVC = 3.33$). Long run: exit ($P < ATC$).
Answer: Shut down: $P <$ min AVC $= 3$. Loss $= 10$, less than producing (e.g. $Q = 2$ loses 11).

Problem: same firm, different prices

Use the cost table from the last slide. Produce every unit whose $MC \leq P$.

1. $P = \$9$: how many units? Profit?
2. $P = \$5$: how many units? Profit? Produce or shut down in the short run? Long run?
3. $P = \$2.50$: what does the firm do?
4. In the long run $P = \$6.50$. Market demand is $Q_D = 1400 - 80P$. How many firms?

Answer: $Q = 5$ (MC of 5th unit $= 8 \leq 9$; 6th $= 10 > 9$). Profit $= 45 - 34 = 11 = (9 - 6.8) \times 5$.
Answer: $Q = 3$. Profit $= 15 - 20 = -5$. Shutting down loses $FC = 10$, so produce ($P = 5 > AVC = 3.33$). Long run: exit ($P < ATC$).
Answer: Shut down: $P <$ min AVC $= 3$. Loss $= 10$, less than producing (e.g. $Q = 2$ loses 11).
Answer: $Q_D = 1400 - 520 = 880$. Firms $= 880/4 = 220$.

Short run vs. long run after a demand increase

Start in long-run equilibrium (P = min ATC, zero profit). Demand rises.

[Pair up] Put these in order:

(a) New firms enter, market supply shifts right.

(b) Price rises above ATC; each firm makes a profit.

(c) Price falls back to min ATC; profit is zero again.

(d) Each existing firm produces more (moves up its MC curve).

[Vote] Compared with the long run, the short-run change in *price* is ___ and the change in *quantity* is ___.

Short run vs. long run after a demand increase

Start in long-run equilibrium (P = min ATC, zero profit). Demand rises.

[Pair up] Put these in order:

(a) New firms enter, market supply shifts right.

(b) Price rises above ATC; each firm makes a profit.

(c) Price falls back to min ATC; profit is zero again.

(d) Each existing firm produces more (moves up its MC curve).

Answer: (b) and (d) happen together in the short run, then (a), then (c).

[Vote] Compared with the long run, the short-run change in *price* is ___ and the change in *quantity* is ___.

Short run vs. long run after a demand increase

Start in long-run equilibrium (P = min ATC, zero profit). Demand rises.

[Pair up] Put these in order:

(a) New firms enter, market supply shifts right.

(b) Price rises above ATC; each firm makes a profit.

(c) Price falls back to min ATC; profit is zero again.

(d) Each existing firm produces more (moves up its MC curve).

Answer: (b) and (d) happen together in the short run, then (a), then (c).

[Vote] Compared with the long run, the short-run change in *price* is ___ and the change in *quantity* is ___.
Answer: Price changes **more** in the short run; quantity changes **less** (2015 exam). Long run: the extra demand is met by **more firms**, not bigger firms (see today's ICA).

Wrap-up

Finish the sentence

1. The socially efficient quantity is where _____ crosses _____.
2. Who specializes in what is decided by _____ advantage.
3. When a country imports, _____ gain and _____ lose.
4. A public good is non-_____ and non-_____.
5. At the consumer's optimum, MRS equals _____.
6. A firm shuts down in the short run if $P <$ _____.
7. In long-run equilibrium, $P =$ _____ and profit = _____.

Finish the sentence

1. The socially efficient quantity is where _____ crosses _____.
2. Who specializes in what is decided by _____ advantage.
3. When a country imports, _____ gain and _____ lose.
4. A public good is non-_____ and non-_____.
5. At the consumer's optimum, MRS equals _____.
6. A firm shuts down in the short run if $P <$ _____.
7. In long-run equilibrium, $P =$ _____ and profit = _____.

Answer: 1. SMB, SMC 2. comparative 3. consumers, producers 4. rival, excludable 5. the price ratio P_X/P_Y 6. AVC 7. min ATC, 0.

Open up Canvas and navigate to **“In-Class Activity 8 (Discussions)”**!

Please scan the QR code and fill out the attendance form!

